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PRIVATE SHIELD: Weekly Implementation Report

I. INTRODUCTION

This week's work focused on studying and attempting to implement four research directions related to privacy-preserving and collaborative intrusion detection for the PRIVATE SHIELD project. A basic end-to-end prototype based on SecureDyn-FL was completed. The other three papers were examined through partial implementation, code planning, and architecture design. No final accuracy or performance claims are reported for the unfinished implementations.

II. IMPLEMENTATION PROGRESS

A. SecureDyn-FL Basic Prototype

The first paper was used to build the completed basic prototype. The implementation included N-BaIoT-compatible data preprocessing, client-wise data division, simulated federated clients, local 1D-CNN training, server-side model averaging, and basic evaluation under IID and non-IID data settings. The complete baseline flow from local training to global-model update was executed successfully. However, the paper's advanced mechanisms, including GMM and Mahalanobis-distance gradient auditing, encrypted aggregation, personalization, pruning, quantization, and full poisoning tests, remain outside the current basic version.

B. WEF-Defense Exploration

The second paper was studied to understand free-rider clients that receive the global model without providing useful local training. The implementation attempt focused on the Weight Evolving Frequency (WEF) concept: recording changes in the penultimate-layer weights, calculating a dynamic threshold, and preparing a WEF-Matrix for distinguishing genuine contributors from suspicious clients. The full client-separation, reputation, and personalized aggregation process has not yet been completed.

C. FedMSE Exploration

For FedMSE, the team reviewed the SAE-CEN anomaly-detection structure and the MSEAvg aggregation method. Initial work concentrated on preparing normal-only N-BaIoT data, defining a Shrink Autoencoder, planning centroid-based anomaly classification, and designing MSE-based weights for local models. The complete multi-gateway simulation, non-IID experiments, and AUC comparison are still pending.

D. Resource-Constrained IDS Adaptation

The fourth source is a review paper rather than one directly reproducible algorithm. Therefore, its findings were converted into implementation requirements for PRIVATE SHIELD. The selected directions include local data processing, federated collaboration, lightweight models, reduced communication, model pruning and quantization, and secure model exchange suitable for low-resource IoT devices.

CURRENT STATUS

Only the SecureDyn-FL basic implementation is currently complete. The WEF-Defense and FedMSE implementations remain at the initial or partial stage, while the review paper has been used mainly to guide system design. Final comparisons cannot be made until all modules run under the same data, client, and evaluation settings.

FUTURE OBJECTIVES

- Complete WEF-Matrix tracking, client reputation, and free-rider separation.
- Finish the SAE-CEN and MSEAvg pipeline with normal-only and non-IID N-BaIoT data.
- Compare accuracy, F1-score, AUC, communication cost, and execution time using one common setup.

RESEARCH SOURCES

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[3] V. T. Nguyen and R. Beuran, “FedMSE: Semi-supervised federated learning approach for IoT network intrusion detection,” *Computers & Security*, vol. 151, Art. no. 104337, Apr. 2025, doi: 10.1016/j.cose.2025.104337.

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